

The intuition behind the Levy–Markowitz approximation in equation (2.5) of mean-variance utility to any expected utility function is that the two most important effects are where the returns are centered (the mean) and how disperse they are (the variance). There is a trade-off between these two effects, which is captured by the investor’s risk aversion. This is precisely the mean-variance utility setup of equation (2.4). Mean-variance utility is ubiquitous in the investment industry (much less so, thankfully, in academia). It has been relentlessly criticized, quite rightly, but mean-variance techniques nonetheless bring us a great deal of intuition that extends to more complex situations. And the two most important things are indeed means and variances. Often other things also matter—and unfortunately mean-variance utility does not allow us to capture anything other than the mean and variance.

Mean-variance utility does not assume returns are normal. Often people confuse using mean-variance utility with assuming normally distributed returns. This error comes about because, with normal distributions, there are only two parameters—the mean and the variance, and these completely describe the normal distribution (they are called *sufficient statistics*). Since with normal distributions only means and variances matter, mean-variance utility is perfect. Levy and Markowitz showed that using mean-variance utility is often a good approximation with non-normal returns. But sometimes the approximation can be lousy. Let’s go back to Figure 2.1 showing the cumulated wealth of the volatility strategy and the S&P 500. Both end up with approximately the same return, and they have approximately the same mean (around 10%) and the same standard deviation (around 15%). But investing in the volatility strategy is like picking up the nickels and dimes before an oncoming steamroller—the skewness is -8 and kurtosis is 104 . The S&P 500 is much closer to a normal distribution. Using only the mean and variance would imply that both the volatility strategy and the S&P 500 would be approximately the same. But clearly the volatility strategy and the S&P 500 are not the same. Would you want to use the mean and variance exclusively?

Having said that, we’ll stick with using means and variances for now and return to more realistic utility functions in section 4.

3.1. INDIFFERENCE CURVES

We represent mean-variance utility pictorially by *indifference curves*. One particular indifference curve represents one particular level of utility. Figure 2.5 plots three different indifference curve for different utility levels for an asset owner with a risk aversion of $\gamma = 3$. We plot these in mean-standard deviation space on the y-axis and x-axis, respectively. Along one particular indifference curve, an investor is indifferent to all the mean-volatility combinations. For example, on the indifference curve with the triangles in Figure 2.5, which corresponds to a utility of $U = 0.15$, the asset owner does not care which mean-volatility combination that she picks; all of them result in the same number of utils. Investors seek the